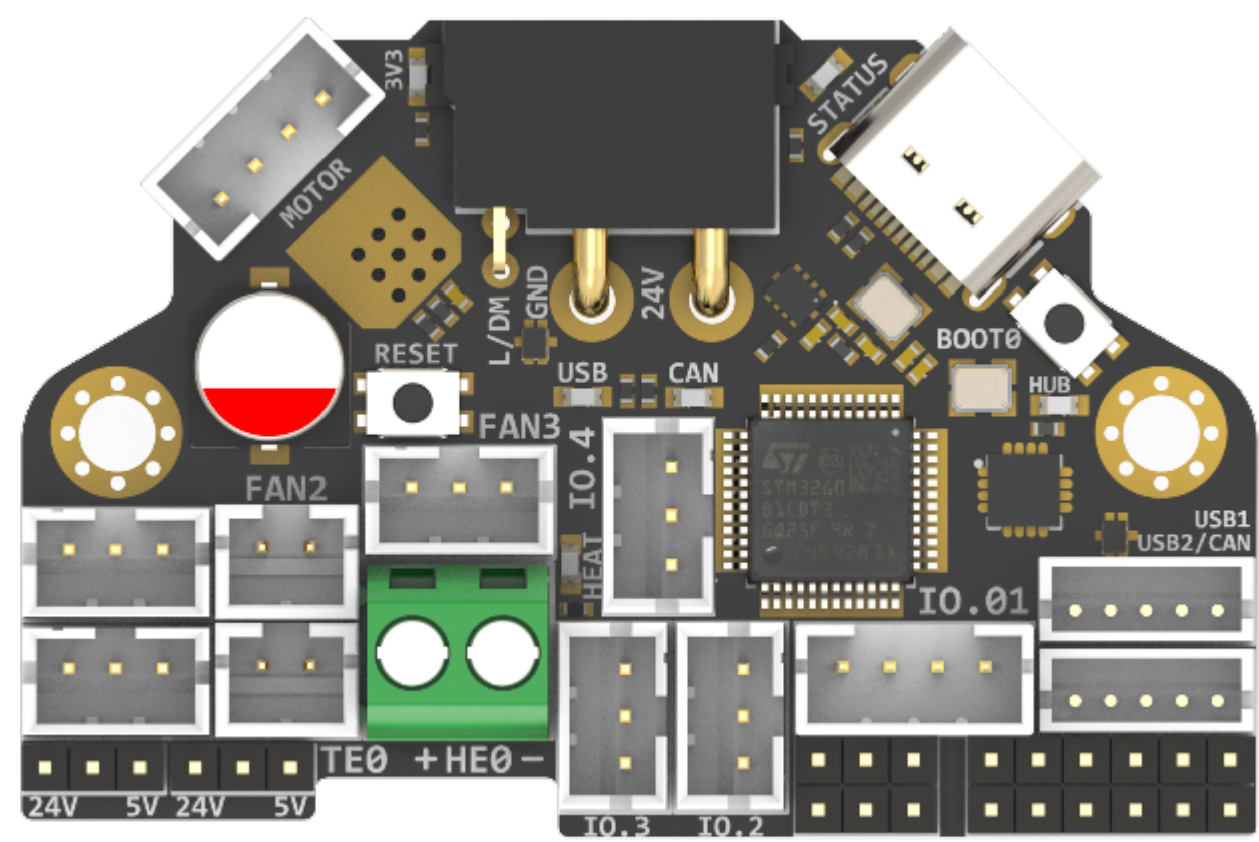
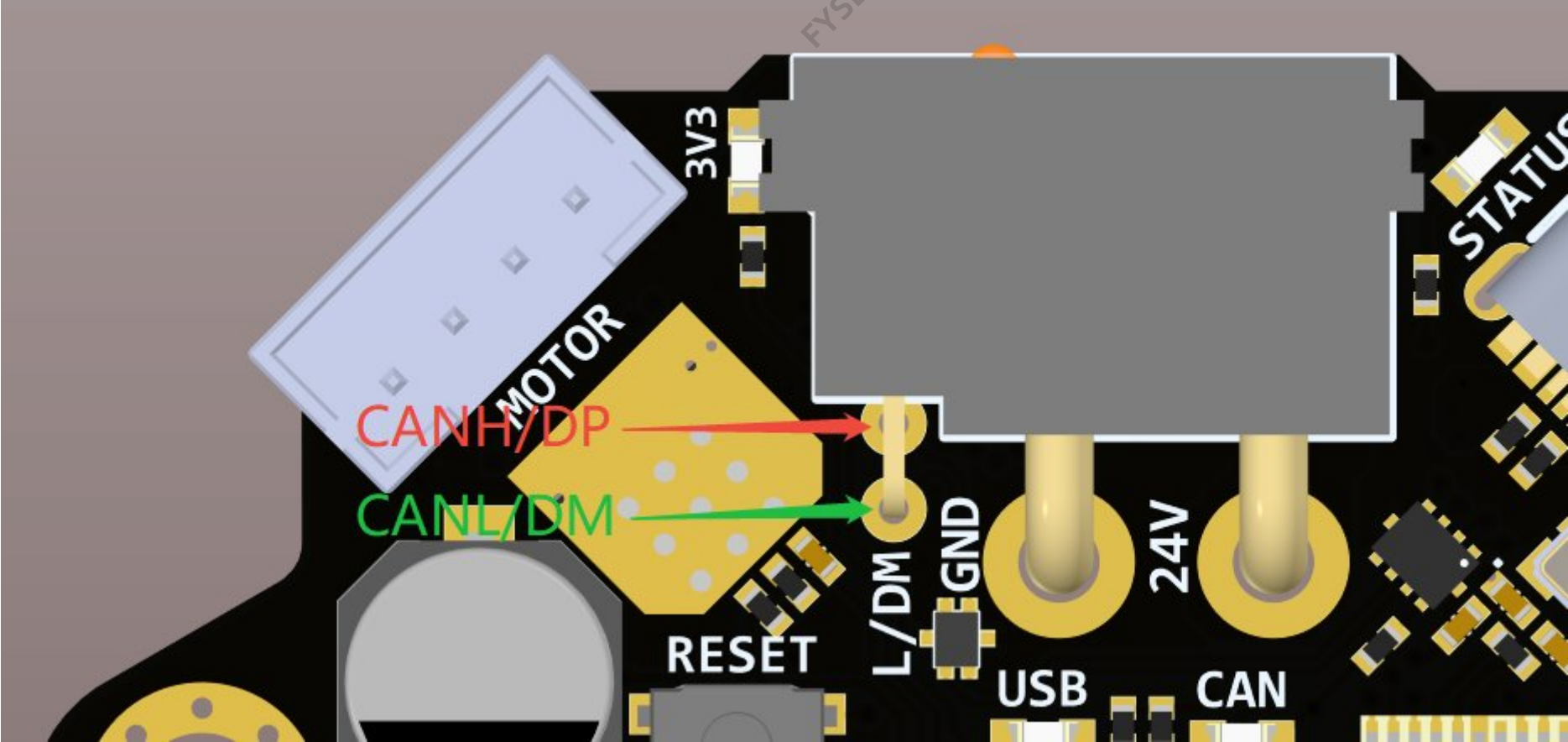


H36 Combo



!!! attention Note

- Since there is no 125 degree Celsius USB-HUB chip, the 125 degree Celsius operating environment is limited to CANBUS mode;
 - Klipper currently does not support 125 degree Celsius accelerometers, so the 125 degree Celsius operating environment does not include accelerometers. The default version of H36 is ADXL345, which has a maximum operating temperature of 85 degrees Celsius. The 105 degree Celsius ADXL345-EP version is optional, but it is more expensive. !!!
- !!! bug Silkscreen Error Sorry for the same silkscreen error on V1.2 and V1.3: The L/DM and H/DP markings on the back of the XT30 connector are reversed, the top layer is correct. To avoid confusion, please refer to the following instructions:



Introduction

H36 Combo V1 is a newly designed **high temperature** tool board running Klipper firmware. Based on STMG0B1T3, the maximum operating temperature can reach **125 degrees Celsius (CAN only)**. It also provides two communication methods: CANBUS and USB. The onboard USBHUB has up to two USB interfaces (or one CAN and one USB), which can be used to connect to the IDM Scanner Leveling Sensor camera, etc.

Version Notes

V1.1

- 1.add Driver NTC
- 2.add 3.3V/5v monitor
- 3.add 24v monitor

V1.2

- 1.change to XT30
- 2.add reserve protection
- 3.fix can/usb indicate

V1.3 (Currently Available) !!! note Note V1.3 only makes hardware changes and does not affect any software configuration. !!!

- 1.Add level_shift to FAN_TACH
- 2.Fix silk mark
- 3.Add jumper resistor from USB-C to STM32
- 4.ADXL345-EP optional
- 5.Change the LDO chip to TLV75733
- 6.New mass-produced cables (Mold Forming)

Featrues

- STM32G0B1T3 MCU, **up to 125 degrees Celsius**, supports Klipper Firmware
- 6-layer PCB design, all solid capacitors and tantalum capacitors, better heat dissipation
- Onboard IO controlled CAN/USB switch,
- Onboard RESET and BOOT0 buttons for easy firmware update
- Fan control/heating both use independent high-power MOS, safer and lower heat generation
- Onboard 5V@3A DC-DC, 5V peak load can reach 15W
- XT30 interface, custom cable comes with the board
- Onboard USB2.0 HUB, 3x USB2.0 interface (one of which is connected to MCU)
- Support CAN / USB2.0 connection
- Onboard TMC2209
- 3x 3pin Fans, 3x IO with level converter, 1x RGB,1x Heat, 1x thermistor,1x status LED
- ADXL345 Accelerometer onboard
- USB-C onboard for Firmware update

Application

Print Head with NEMA14/36mm Motor

Hardware specification

Function	H36 Combo V1	SB Combo V2	SB TH CAN V1.3
Microcontroller	STM32G0B1T3	STM32F072CBT6	STM32F072CBT6
Accelerometer	ADXL345	ADXL345	ADXL345
USB Port	2 + 1 (For MCU) , Powered by CH334P	2 + 1 (For MCU) + 1(on aux board), Powered by CH334P	1
CANBUS	1 + 1 (For MCU)	1 + 1 (For MCU)	1
Fan	4 (1x2Pin, 3x3Pin)	3+2 (on aux board)	2
Heating output	1	1	1
Temperature measurement	1 (PH2.0 connector) + 1 (on board thermistor)	1 (PH2.0 connector) + 1 (on board thermistor)	1(PH2.0 connector)
Voltage monitoring	5V, 24V	5V, 24V	-
Motor drive	TMC2209	TMC2209	TMC2209
Signal input and output	5 (PH2.0) + 7 (pin header)	3	3

Function	H36 Combo V1	SB Combo V2	SB TH CAN V1.3
RGB light strip control	default 2, up to 5	1 + 1 (on aux board)	1
Aux Board support	No	Yes	No
Onboard mounting nut	No	Yes	No

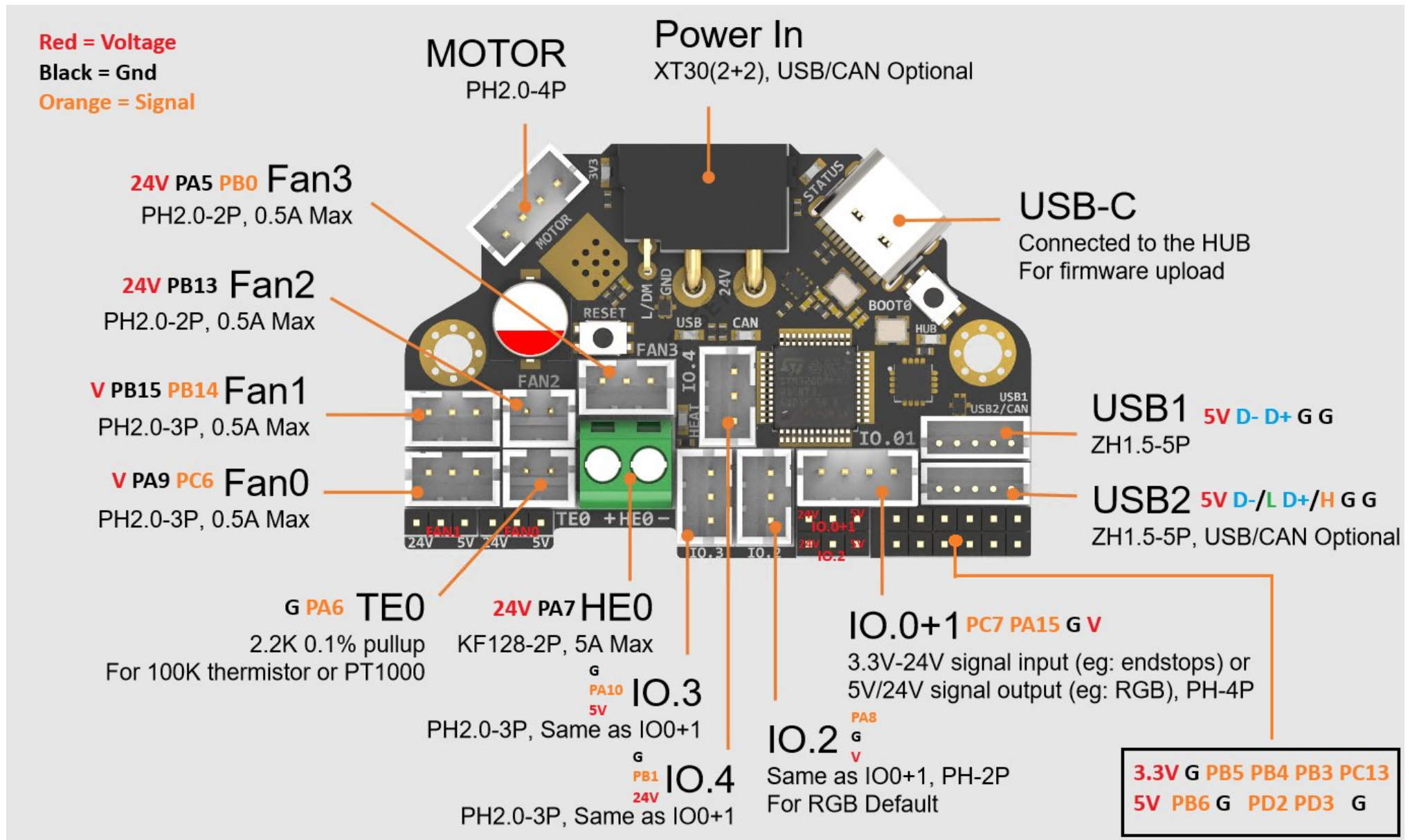
Operating limits

!!! warning Waining The following values are tested at room temperature of 25°C. Please do not keep the highest peak value running for a long time! Please do a good job of cooling the board in a higher temperature environment. !!!

Stepper drivers	Up to 2.0A peak current
Mosfets Outputs	HE0 up to 5A Max, Fan up to 0.5A each
Input power voltage	11V to 24V for VIN up to 10A Max
Inputs/Outputs	Signal 20ma maximum, RGB power supply 1.5A total maximum
5V and 3.3V current limit	5V@3A Max,3.3V@0.8A Max

Physical Connections

Pin Out



Thanks to Esoterical for creating this image.

IO.0: PA15
IO.1: PC7
IO.2: PA8
IO.3/RGB0: PA10
IO.4/RGB1: PB1
IO.0 - IO.5 with level conversion (pull-up), it can be used as 5V/24V level input and output!

FAN0: PA9
FAN0_TACH: PC6 # Please make sure the feedback level does not exceed 5V!
FAN1: PB15
FAN1_TACH: PB14 # Please make sure the feedback level does not exceed 5V!
FAN2: PB13
FAN3: PA5

FAN3_TACH: PB0 # Please make sure the feedback level does not exceed 5V!

Heat: PA7

Thermistor (2.2k pull up): PA6

Driver_EN: PB7

Driver_DIR: PB8

Driver_STEP: PB9

Driver_UART: PC14

Driver_DIAG: PC15

ADXL345

spi_bus: spi2

adxl345_cs_pin: PB12

adxl345_spi_software_sclk_pin: PB10

adxl345_spi_software_mosi_pin: PB11

adxl345_spi_software_miso_pin: PB2

CAN_RX: PD0

CAN_TX: PD1

5V_Monitor: PA0

24V_Monitor: PA4

Thermistor next to TMC2209 (4.7k pull up) : PA3

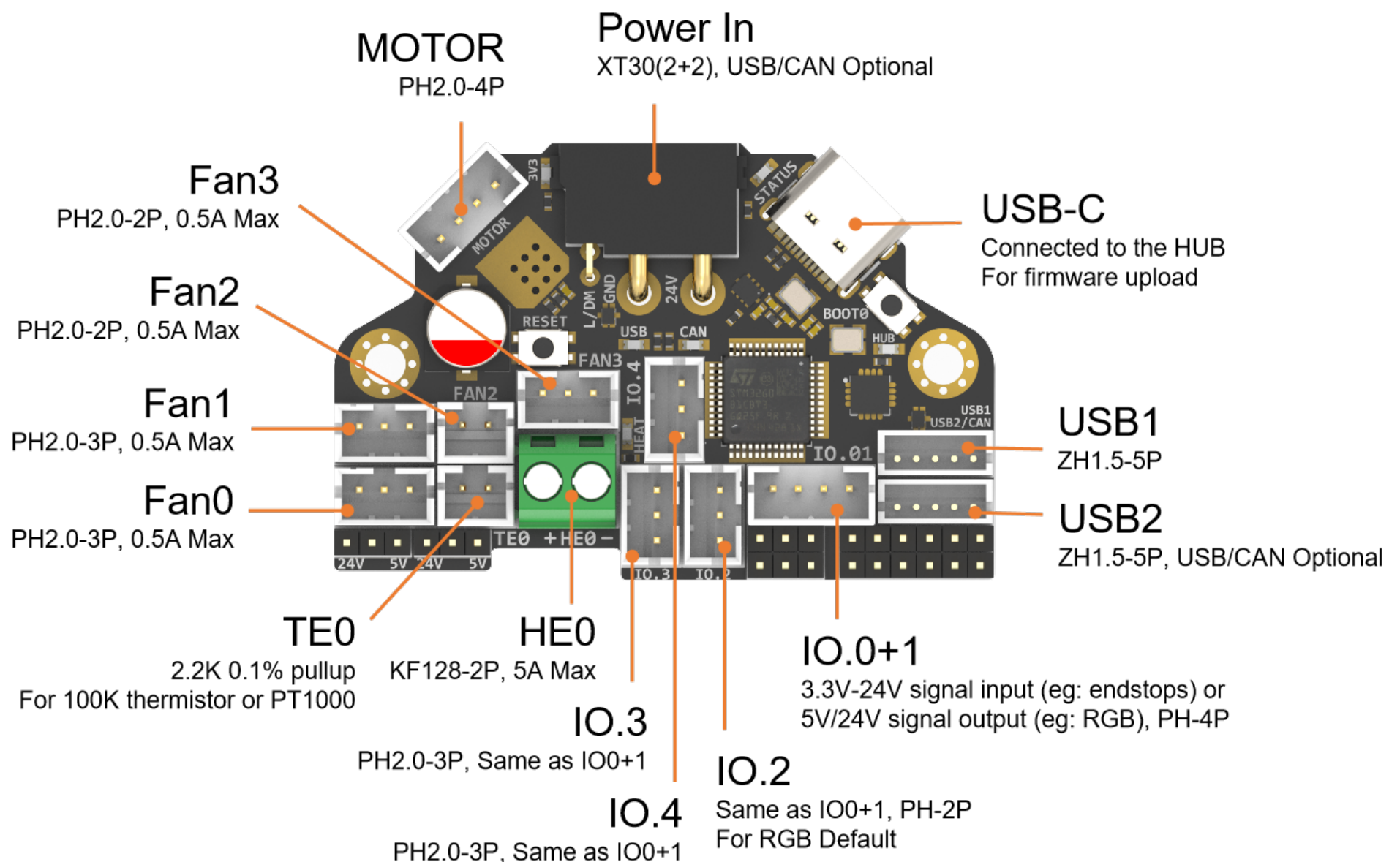
CAN USB switch: PA2

logic Low = COM to A PORT CANBUS

Logic High = COM to B PORT USB

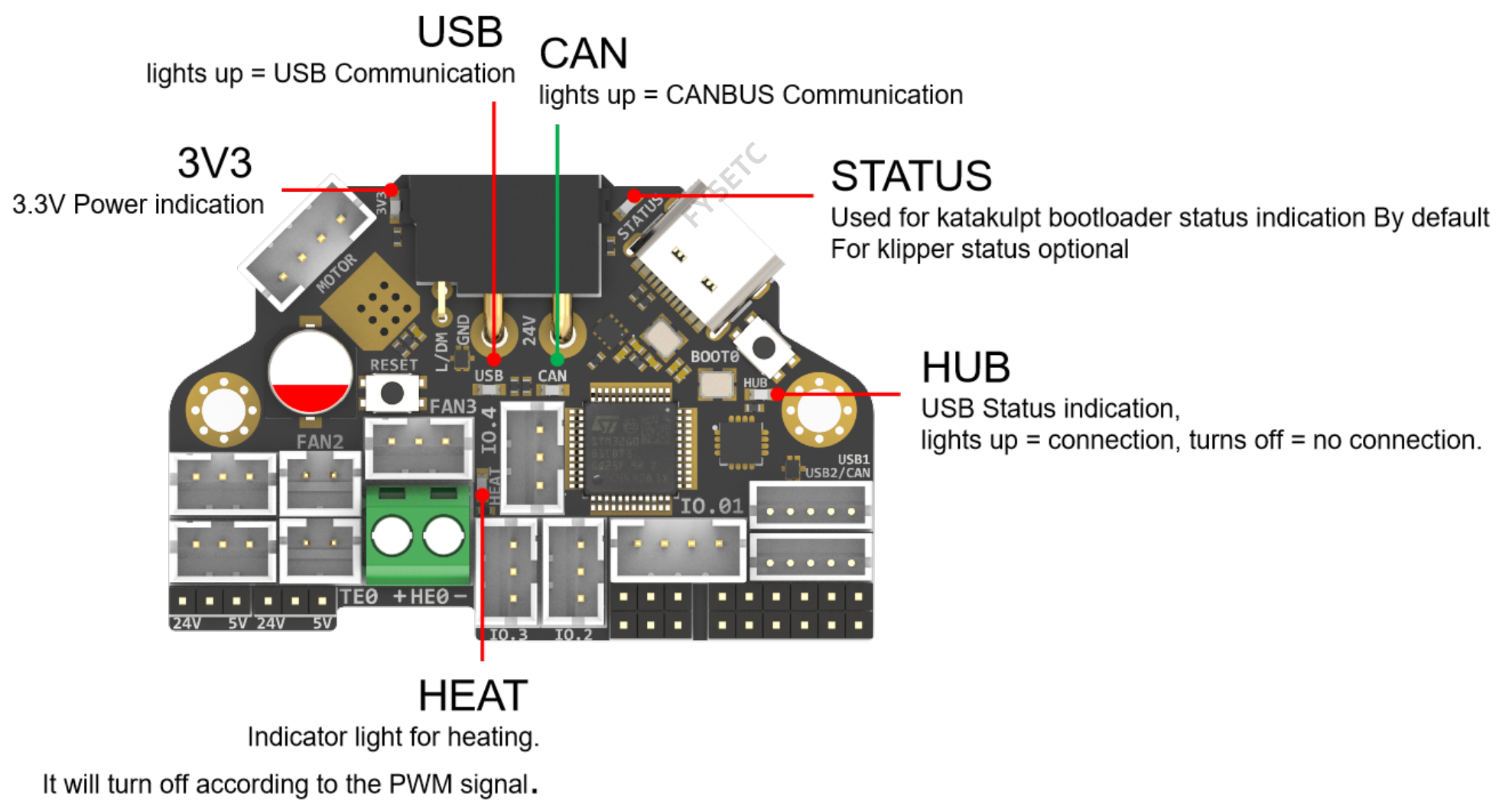
Description of Connections:

IO.0 and IO.1 are located in the same connector and can be used as XY endstops, and IO.2 can be used as a probe. In addition, IO.0 - IO.5 have level conversion (with pull-up resistors), IO.01, IO.2 have voltage selectors (5V or 24V) pin header, IO.3 IO.4 have voltage selectors (5V or 24V) pads. All the IOs can be used as inputs or outputs, such as RGB ,Servo, Endstops, and can be configured according to your other needs.



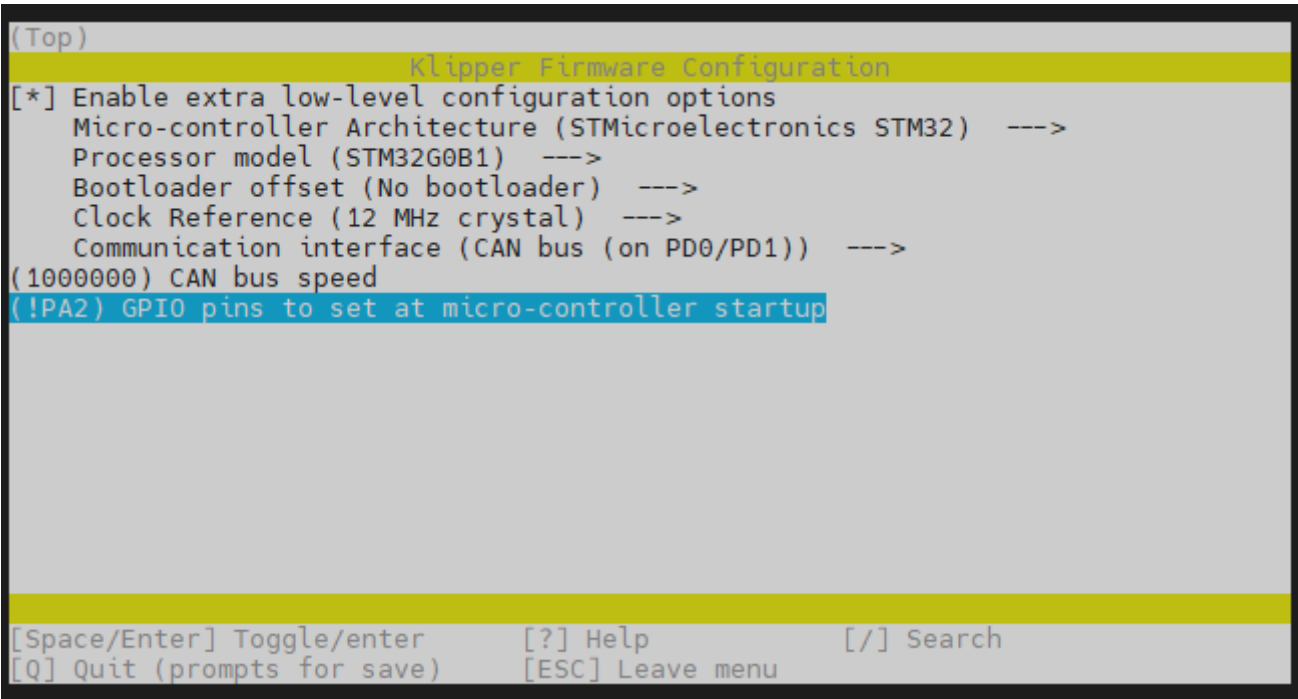
Connector	Pin	Default function	Alternate
Power Input	USB (PA11,PA12) CANBUS (PD0,PD1)	Power and communication input, USB 2.0 and CANBUS are optional, Determined by the level of PA2.	
Fan0	FAN0: PA9 FAN0_TACH: PC6	Mosfet Output, For 2/3 pins Fan, Default voltage = Depending on voltage selector. (Header voltage selector)	
Fan1	FAN1: PB15 FAN1_TACH: PB14	Mosfet Output, For 2/3 pins Fan, Default voltage = Depending on voltage selector. (Header voltage selector)	
Fan2	PB13	Mosfet Output, For 2 pins Fan, Default voltage = VIN. (Pad voltage selector)	
Fan3	FAN3: PA5 FAN3_TACH: PB0	Mosfet Output, For 2/3 pins Fan, Default voltage = VIN. (Pad voltage selector)	
IO.0+1	PA15, PB15	Digital Input, For X Y endstop, Micro switch or Hall (Header voltage selector)	Digital Output
IO.2	PA8	Digital Input, For Z probe, Proximity switch or Klicky, etc. (Header voltage selector)	Digital Output
IO.3/RGB0	PA10	5V Digital Output, 5V power supply default, For 5V WS2812/SK6812 RGB/Servo (Pad voltage selector)	Digital Input
IO.4/RGB1	PB1	5V Digital Output, 24V power supply default, For 24V WS2812/SK6812 RGB/Servo (Pad voltage selector)	Digital Input
USB1		USB2.0	J2
USB2/CAN		USB2.0 or CANBUS 2.0, USB 2.0 and CANBUS are optional,Determined by the level of PA2.	J1
TE0	PA6	ADC input, 2.2K pull-up, head temperature measure	
HE0	PA7	Mosfet Output, Heating rod control, 5A Max	
MOTOR		For two-phase stepper motor,	
USB1, USB-C connector		Connected to the CH334PHUB chip, up to 4 USB2.0 devices (MCU/USB1/USB2/USB3)	

LED indications:

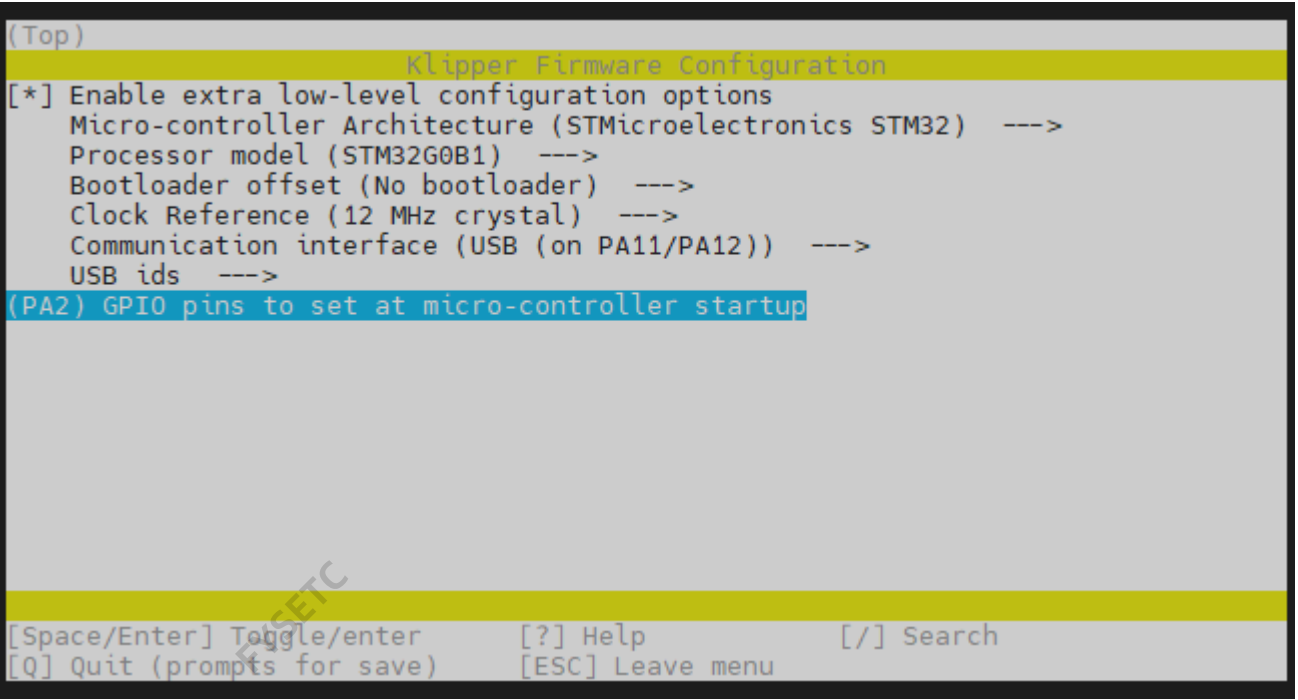


LED name	Indicate	Remark
3V3	Lights up: Power supply OK. Turns off : Power supply failure. 3.3V is obtained by converting 24V to 5V through DC-DC and then to 3.3V through LDO, so there may be a short circuit/open circuit in 24V/5V/3.3V.	Red
HUB	Lights up: The USB has at least one connection. Turns off: The USB has no connection.	Red
STATUS	When using katakulpt- Blinking: Entering download mode; Off/Always on: Not in download mode, the status is generally determined by config; The LED is controlled by PC13 and lights up at a high level. For customized usage, please refer to julianschill/klipper-led_effect	Red
HEAT	Lights up or flashes according to the heating PWM	Red
USB	USB communication Indicate, use "PA2" at the config menu	Red
CAN	CAN communication Indicate, use "!PA2" at the config menu	Green

USB / CAN selection



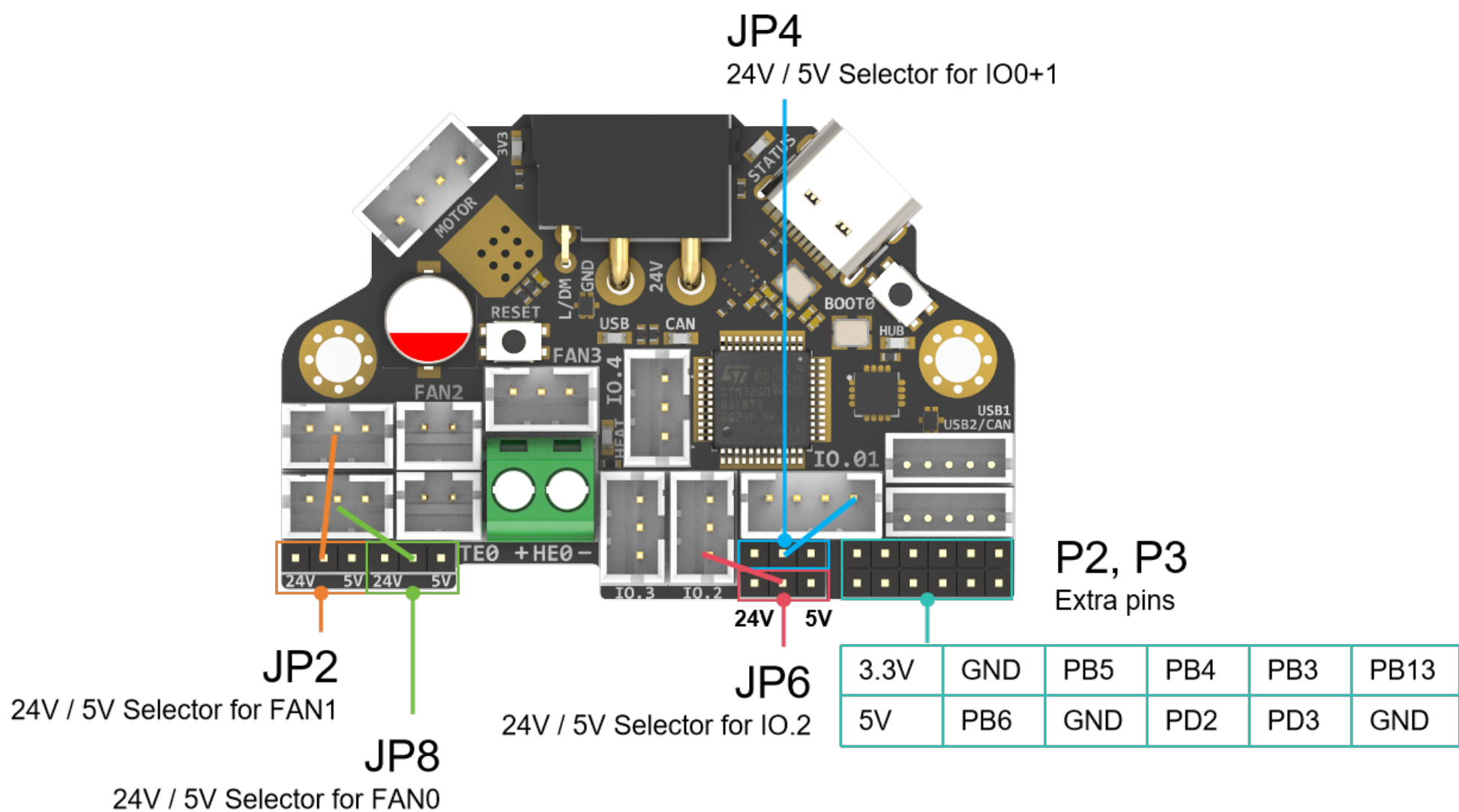
XT30 and USB2/CAN both use CAN to communicate



XT30 and USB2/CAN both use USB to communicate

Header Jumpers:

IO.0+1 , IO.2 , Fan0 , Fan1 can select the power supply voltage through the jumper cap. As shown in the figure, the two pins on the left are connected together for 24V, and the two pins on the right are connected together for 5V.



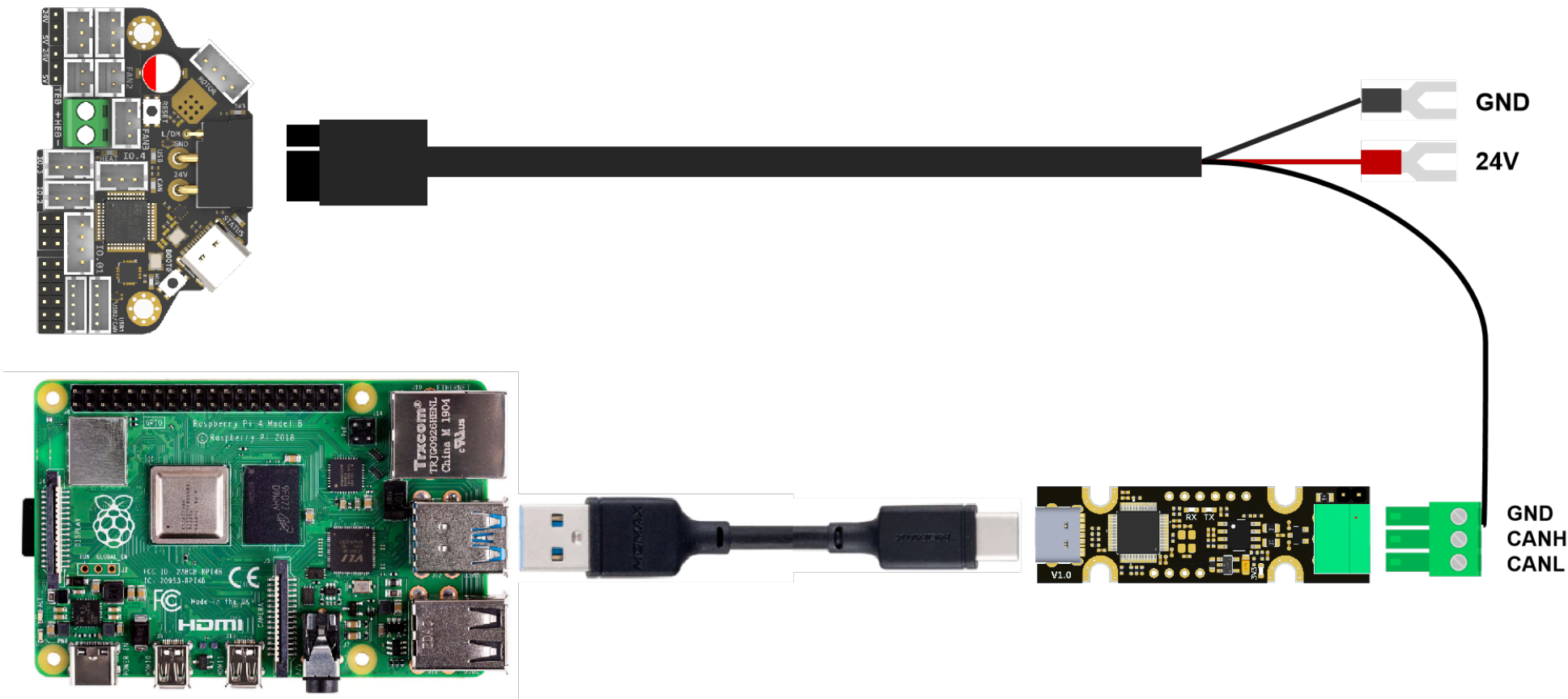
!!! note NOTE Please note that if these interfaces are used as outputs, the high-level voltage of the output is consistent with the voltage selected by the jumper. Please make sure that your peripherals can withstand the range. Generally speaking, only SSR in the accessories of 3D printers can withstand 9-36V control voltage. !!!

Communication

Connect to Pi

SB combo V2 can be connected to Pi via USB or CANBUS. Select by PA2, refer to USB / CAN selection.

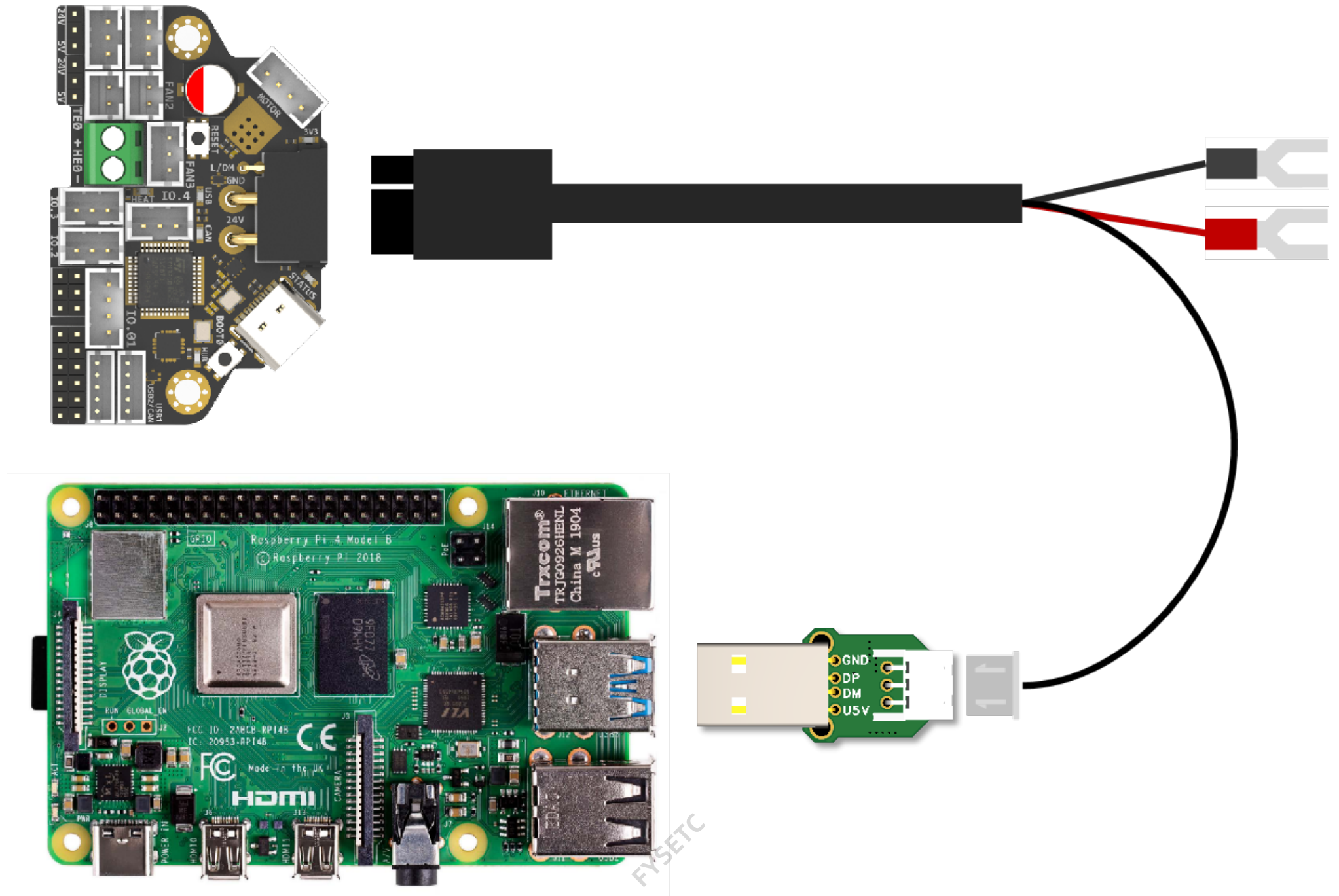
Via CAN



Since the Raspberry Pi does not have a CANBUS interface, it is usually connected only after the CANBUS interface is expanded through an expansion chip. Commonly used ones include the MCP2518 SPI to CAN module (CanHat), USB to CAN module (eg: UCAN), Klipper USB to CAN Bridge Mode, and a Linux host computer with a native CAN interface (eg: CM68).

Via USB

When using USB connection, you can use the adapter board we provide to directly connect to the USB interface of PI.



Firmware Guide

Firmware configuration and compilation

With Katapult Bootloader

If you need to use a bootloader, we recommend using katapult, the following is not configured for reference: For katapult use, refer to: <https://github.com/Arksine/katapult>

```
cd ~/katapult
make menuconfig
make flash FLASH_DEVICE=0483:df11
```



```
(Top)
Katapult Configuration v0.0.1-64-g3e23332-dirty
Micro-controller Architecture (STMicroelectronics STM32) --->
Processor model (STM32G0B1) --->
Build Katapult deployment application (Do not build) --->
Clock Reference (12 MHz crystal) --->
Communication interface (USB (on PA11/PA12)) --->
Application start offset (8KiB offset) --->
USB ids --->
(PA2) GPIO pins to set on bootloader entry
[*] Support bootloader entry on rapid double click of reset button
[ ] Enable bootloader entry on button (or gpio) state
[*] Enable Status LED
(PA1) Status LED GPIO Pin

[Space/Enter] Toggle/enter      [?] Help      [/] Search
[Q] Quit (prompts for save)     [ESC] Leave menu
```

```
(Top)
Katapult Configuration v0.0.1-64-g3e23332-dirty
Micro-controller Architecture (STMicroelectronics STM32) --->
Processor model (STM32G0B1) --->
Build Katapult deployment application (Do not build) --->
Clock Reference (12 MHz crystal) --->
Communication interface (CAN bus (on PD0/PD1)) --->
Application start offset (8KiB offset) --->
(1000000) CAN bus speed
(!PA2) GPIO pins to set on bootloader entry
[*] Support bootloader entry on rapid double click of reset button
[ ] Enable bootloader entry on button (or gpio) state
[*] Enable Status LED
(PA1) Status LED GPIO Pin

[Space/Enter] Toggle/enter      [?] Help      [/] Search
[Q] Quit (prompts for save)     [ESC] Leave menu
```

```
cd ~/klipper
make menuconfig
make
cd ~/katapult/scripts
#when use CAN communication
python3 flashtool.py -i can0 -f ~/klipper/out/klipper.bin -u <uuid>
#when use USB communication
python3 flashtool.py -d <serial device> -b <baud_rate>
```

```
(Top)
Klipper Firmware Configuration
[*] Enable extra low-level configuration options
Micro-controller Architecture (STMicroelectronics STM32) --->
Processor model (STM32G0B1) --->
Bootloader offset (8KiB bootloader) --->
Clock Reference (12 MHz crystal) --->
Communication interface (CAN bus (on PD0/PD1)) --->
(1000000) CAN bus speed
(!PA2) GPIO pins to set at micro-controller startup

[Space/Enter] Toggle/enter      [?] Help      [/] Search
[Q] Quit (prompts for save)     [ESC] Leave menu
```

```
(Top)
Klipper Firmware Configuration
[*] Enable extra low-level configuration options
  Micro-controller Architecture (STMicroelectronics STM32) --->
  Processor model (STM32G0B1) --->
  Bootloader offset (8KiB bootloader) --->
  Clock Reference (12 MHz crystal) --->
  Communication interface (USB (on PA11/PA12)) --->
  USB ids --->
(PA2) GPIO pins to set at micro-controller startup

[Space/Enter] Toggle/enter    [?] Help    [/] Search
[Q] Quit (prompts for save)  [ESC] Leave menu
```

Klipper with nobootloader

H36 uses the bootloader-free mode by default, and the configuration is as follows:

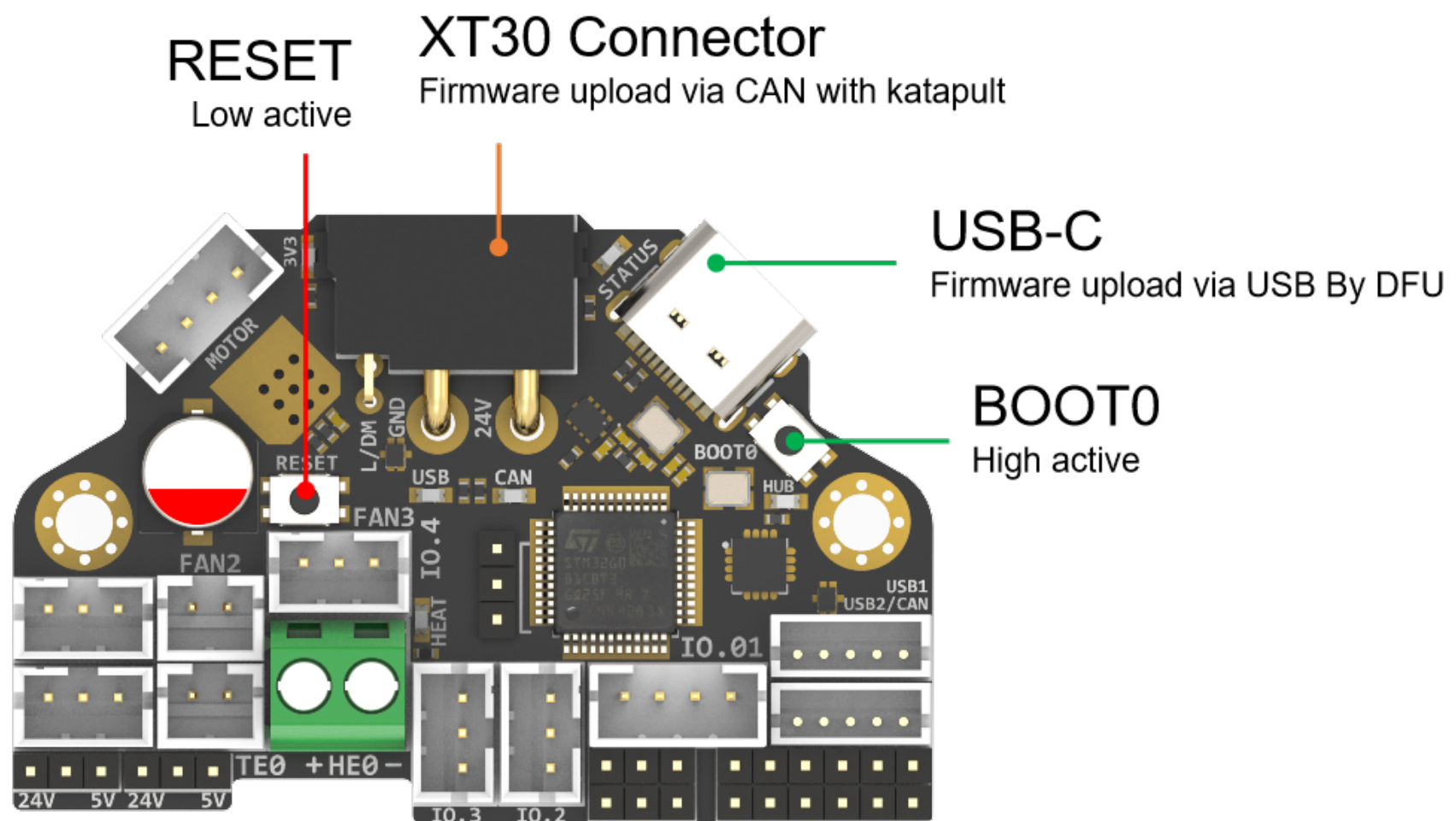
```
(Top)
Klipper Firmware Configuration
[*] Enable extra low-level configuration options
  Micro-controller Architecture (STMicroelectronics STM32) --->
  Processor model (STM32G0B1) --->
  Bootloader offset (No bootloader) --->
  Clock Reference (12 MHz crystal) --->
  Communication interface (CAN bus (on PD0/PD1)) --->
(1000000) CAN bus speed
(!PA2) GPIO pins to set at micro-controller startup

[Space/Enter] Toggle/enter    [?] Help    [/] Search
[Q] Quit (prompts for save)  [ESC] Leave menu
```

```
(Top)
Klipper Firmware Configuration
[*] Enable extra low-level configuration options
  Micro-controller Architecture (STMicroelectronics STM32) --->
  Processor model (STM32G0B1) --->
  Bootloader offset (No bootloader) --->
  Clock Reference (12 MHz crystal) --->
  Communication interface (USB (on PA11/PA12)) --->
  USB ids --->
(PA2) GPIO pins to set at micro-controller startup

[Space/Enter] Toggle/enter    [?] Help    [/] Search
[Q] Quit (prompts for save)  [ESC] Leave menu
```

Firmware upload



Before executing the following commands, you need to enter DFU mode before you can compile and burn the firmware. As shown in the figure above,

- press and hold BOOT0,
- then press the RESET button for one second,
- then release RESET,
- wait for 3 seconds,
- and release BOOT0.

Use "lsusb" to check if a DFU device appears. If yes, upload firmware. If no, repeat the above steps.

make flash FLASH_DEVICE=0483:df11

under katapult Klipper uploading, Please refer to:

<https://github.com/Arksine/katapult?tab=readme-ov-file#uploading-klipper>

Attachments and others

2D, 3D, SCH and config template file, please go to our github:

https://github.com/FYSETC/H36_Combo

FAQ

1. Why I can't enter the DFU mode?

There are two reasons:

1. The operation of boot0 and reset buttons may be incorrect, or the buttons are damaged. Solution:
 - A. Repeat the above tutorial;
 - B. Unplug all the cables of the board, press BOOT0, insert USB-C, wait for a few seconds, and the computer/PI will recognize the DFU device
 - C. If the above method does not work, use the multimeter beep position to measure the BOOT0 and RESET buttons to ensure that they are in different states when pressed and released. If they are in the same state, the buttons are physically damaged, please contact the place of purchase for replacement.
2. STM32G0B1 is different from other STM32s. Its mode of entering DFU is set to software operation at the factory. We need to change its BOOTSEL configuration bit during production and change it to enter DFU through hardware switching, that is, key. If you encounter a situation where the key operation cannot enter DFU, then this step may have been missed during factory production. The solution is to use STLINK/Jlink to configure this BOOTSEL bit (If you don't have STlink/Jlink or have no way to solve this problem, you can contact our customer

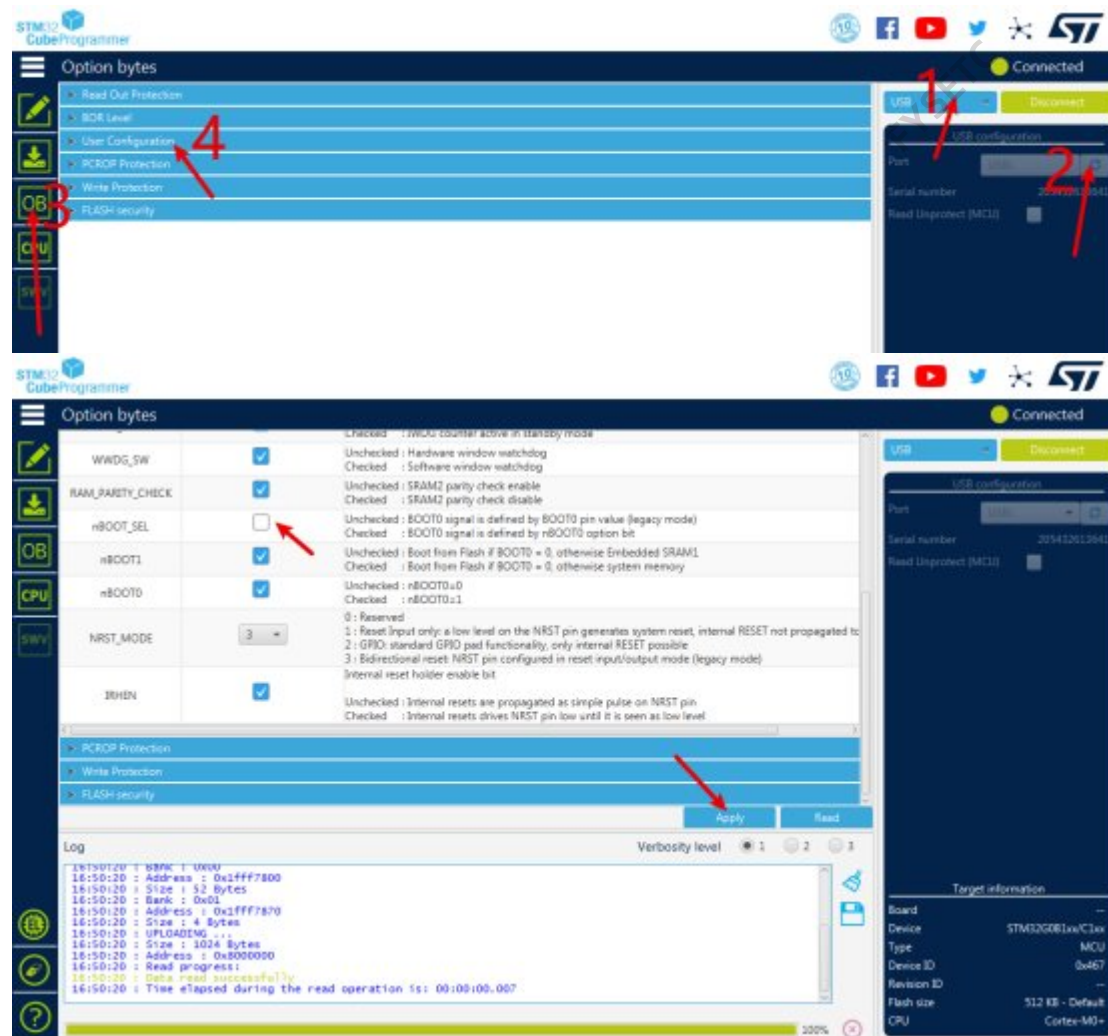
service for help!). The specific operation is as follows:

- A. Prepare an STLINK/Jlink. The following takes STLINK as an example. JLINK is similar;
- B. Connect STLINK through the two test points on the back of H36, CLK and DIO, and any GND and 3.3V/5V power supply on the board: CLK <-----> SWCLK DIO <-----> SWDIO 3.3V/5V <-----> 3.3V/5V GND <-----> GND
- C. Connect STLINK to the computer and set STLINK as follows: (STM32CubeProgrammer software is used here, you can download it from ST



official website)

- D. Click CONNECT, you will see your G0B1 device in area B. If any error is prompted, please return to check the wiring sequence and power supply;
- E. Click the "OB" icon ----> User Configuration, scroll down to find the "nBOOT_SEL" option, click the box on the right, and cancel the selection here, that is, make sure there is no "check mark" here;



- F. Click the "Apply" button below to apply the setting;
- G. To ensure correct operation, it is recommended to unplug the USB cable of STLINK at the computer, plug it in again, and repeat the above operation to ensure that the "nBOOT_SEL" option has been correctly changed. Through the above operation, G0B1 can enter DFU by pressing the button.

How to buy

Technical Support Channel

support@fysetc.com

<https://discord.gg/4A5QDCYx>

FYSETC